5D Action to Toy Functional Mapping

For Z_N Anisotropy Normalization

EDC Project

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Abstract

We establish the explicit mapping from the 5D brane-world action $S_{5D} = S_{\text{bulk}} + S_{\text{brane}} + S_{\text{GHY}}$ to the toy functional $E[u] = (T/2) \int (u')^2 d\theta + \lambda \sum_n W(u(\theta_n))$ used in the Z_N anisotropy normalization derivation. The mapping identifies the tension parameter T, anchor coupling λ, anchor potential W, and angular coordinate θ. All steps are tagged with epistemic status.

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1 Overview: The Mapping Chain

1.1 Starting Points

5D Action (established in EDC):

$$\boxed{S_{5D} = S_{\text{bulk}} + S_{\text{brane}} + S_{\text{GHY}}} \tag{1}$$

where (following Companion C notation):

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g}\,(R - 2\Lambda_5) \tag{2}$$

$$S_{\text{brane}} = -\sigma \int d^4\xi \sqrt{-h} + S_{\text{matter}} \tag{3}$$

$$S_{\text{GHY}} = \frac{1}{\kappa_5^2} \oint d^4\xi \sqrt{-h}\, K \tag{4}$$

Toy Functional (target):

$$\boxed{E[u] = \frac{T}{2} \int_0^{2\pi} (u')^2\, d\theta + \lambda \sum_{n=0}^{N-1} W(u(\theta_n))} \tag{5}$$

1.2 The Reduction Chain

$$\boxed{\underbrace{S_{5D}}_{\text{5D action}} \xrightarrow[\mathrm{Z}_N \text{ ansatz}]{\text{reduction}} \underbrace{S_{4D}[u]}_{\text{4D on ring}} \xrightarrow[\text{static}]{\text{integration}} \underbrace{E[u]}_{\text{1D functional}}} \tag{6}$$

2 Stage 1: Geometry Setup

2.1 Z_N Ring Embedding

Postulate 2.1 (Ring Geometry)**.** *[P] Consider a circular submanifold $\mathcal{R} \subset \Sigma^4$ parameterized by angular coordinate $\theta \in [0, 2\pi)$.*

Definition 2.2 (Angular Coordinate)**.** [P] The coordinate θ is defined via the transverse embedding such that:

- The Z_N fixed points are located at $\theta_n = 2\pi n/N$
- The metric on $\mathcal{R}$ is $ds^2_{\mathcal{R}} = R^2 d\theta^2$
- R is the characteristic ring radius

Postulate 2.3 (Field Variable)**.** *[P] Let $u(\theta)$ be a scalar field on the ring representing:*

- ***Option A:*** *Brane displacement in the ξ-direction (transverse excursion)*
- ***Option B:*** *Mode envelope of a fermion zero mode*
- ***Option C:*** *Thickness variation of the membrane*

We proceed with the general $u(\theta)$ without committing to a specific identification.

2.2 Z_N Symmetry Constraint

Definition 2.4 (Z_N Symmetry)**.** **[Der]** The field $u(\theta)$ has Z_N symmetry:

$$u\left(\theta + \frac{2\pi}{N}\right) = u(\theta) \tag{7}$$

Proposition 2.5 (Fourier Restriction)**.** ***[Der]*** *Under Z_N symmetry, only modes with period $2\pi/N$ survive:*

$$u(\theta) = u_0 + \sum_{m=1}^{\infty} a_m \cos(mN\theta) \tag{8}$$

3 Stage 2: Brane Action Reduction

3.1 Brane Tension Contribution

Postulate 3.1 (Brane Action Ansatz)**.** ***[P]*** *The brane action on the ring takes the form:*

$$S_{brane}[on\ \mathcal{R}] = -\sigma \int d\theta \sqrt{h_{\theta\theta}} \tag{9}$$

where $h_{\theta\theta}$ is the induced metric component.

Proposition 3.2 (Induced Metric with Deformation)**.** ***[Dc]*** *For a brane with transverse displacement $u(\theta)$:*

$$h_{\theta\theta} = R^2 + (u')^2 \tag{10}$$

where $u' = du/d\theta$.

Proof. From the embedding $X^A(\theta) = (\theta, u(\theta))$, the tangent vector is $e^A = (1, u')$ and the induced metric is $h_{\theta\theta} = g_{AB}e^A e^B = g_{\theta\theta} + g_{uu}(u')^2 = R^2 + (u')^2$. □

3.2 Gradient Energy Extraction

Theorem 3.3 (Gradient Energy Term)**.** ***[Dc]*** *Expanding to quadratic order in u':*

$$\sqrt{h_{\theta\theta}} = R\sqrt{1 + (u'/R)^2} \approx R + \frac{(u')^2}{2R} \tag{11}$$

The brane action becomes:

$$S_{brane} \approx -\sigma R \cdot 2\pi - \frac{\sigma}{2R}\int_0^{2\pi} (u')^2\, d\theta \tag{12}$$

Dropping the constant, the ***gradient energy*** *is:*

$$\boxed{E_{grad} = \frac{\sigma}{2R}\int_0^{2\pi} (u')^2\, d\theta} \tag{13}$$

Mapping: Tension Parameter

$$\boxed{T = \frac{\sigma}{R}} \tag{14}$$

5D origin: T is the ratio of brane tension σ to ring radius R.
Epistemic status: [Dc]— derived under small-deformation approximation.

4 Stage 3: Israel Junction Conditions at Z_N Points

4.1 Localized Sources at Fixed Points

Postulate 4.1 (Discrete Defects). ***[P]*** *At each Z_N fixed point θ_n, there exists a localized contribution to the stress-energy:*

$$\tau_{\mu\nu}^{defect} = \sum_{n=0}^{N-1} \tau_n \, h_{\mu\nu} \, \delta(\theta - \theta_n) \tag{15}$$

Theorem 4.2 (Israel Condition at Fixed Points). ***[Der]*** *The Israel junction condition [**?**]:*

$$[K_{\mu\nu}] - h_{\mu\nu}[K] = -\kappa_5^2 S_{\mu\nu} \tag{16}$$

At a Z_N fixed point with localized defect stress τ_n:

$$[K]_{\theta_n} = -\kappa_5^2 \tau_n \tag{17}$$

4.2 Anchor Energy from Localized Extrinsic Curvature

Postulate 4.3 (Extrinsic Curvature Coupling). ***[Dc]*** *The extrinsic curvature at θ_n depends on the field value:*

$$K(\theta_n) = K_0 + K_1 \cdot u(\theta_n) + O(u^2) \tag{18}$$

where K_0, K_1 are geometric coefficients.

Theorem 4.4 (Discrete Anchor Energy). ***[Dc]*** *Integrating the GHY + defect contributions over the N fixed points:*

$$E_{disc} = \sum_{n=0}^{N-1} \left[\frac{1}{\kappa_5^2} K(\theta_n) + \tau_n u(\theta_n) \right] \tag{19}$$

Defining $\lambda \equiv \tau_n$ (identical defects) and $W(u) \equiv u + const$:

$$\boxed{E_{disc} = \lambda \sum_{n=0}^{N-1} W(u(\theta_n))} \tag{20}$$

Mapping: Anchor Coupling

$$\boxed{\lambda = \tau_n = \kappa_5^2 \cdot (\text{defect stress per anchor})} \tag{21}$$

5D origin: λ is the localized stress-energy at each Z_N fixed point, arising from topological defects (Y-junctions, vortices, or pinning sites).
Key assumption: All N anchors have *identical* $\tau_n = \lambda$.
Epistemic status: [Dc]— assumes Z_N symmetry of defect configuration.

5 Stage 4: Combined Functional

Theorem 5.1 (Total Energy Functional). ***[Dc]*** *Combining Eq.* (13) *and Eq.* (20)*:*

$$\boxed{E[u] = \frac{T}{2} \int_0^{2\pi} (u')^2 \, d\theta + \lambda \sum_{n=0}^{N-1} W(u(\theta_n))} \tag{22}$$

This is exactly the toy functional used in the anisotropy normalization derivation.

6 Mapping Dictionary

5D → Toy Mapping Dictionary

Toy Symbol	5D Origin	Formula	Status
θ	Angular coordinate on ring	$\theta \in [0, 2\pi)$	[P]
$u(\theta)$	Brane displacement / mode envelope	—	[P]
T	Brane tension / ring radius	$T = \sigma/R$	[Dc]
λ	Defect stress at $\mathbb{Z}_N$ fixed points	$\lambda = \kappa_5^2 \tau_n$	[Dc]
$W(u)$	Extrinsic curvature coupling	$W(u) \approx u$	[Dc]
θ_n	$\mathbb{Z}_N$ fixed points	$\theta_n = 2\pi n/N$	[Der]
N	Order of discrete symmetry	$\lvert Z_N \rvert$	[Der]

7 The Complete Chain

Complete Derivation Chain: 5D → $a/c = 1/N$

ASCII Diagram:

```
S_5D = S_bulk + S_brane + S_GHY
              |
              | [Dc] Ring embedding, Z_N symmetry
              v
S_4D[u] on ring R
              |
              | [Dc] Small deformation, static limit
              v
E[u] = (T/2)  (u')^2 d  +   Σ_n W(u( _n))
              |
              | [Der] Fourier expansion, Euler-Lagrange
              v
u( ) = u_0 + a_1 cos(N ),  a_1   1/N
              |
              | [Der] Identification c = u_0, a = a_1
              v
a/c = 1/N  (equal corner share)
              |
              | [Der] Averaging ratio formula
              v
k(N) = 1 + 1/N
```

8 Epistemic Status Summary

Step	Status	What Would Upgrade to [Der]
5D action structure	[P]	Derive from Plenum dynamics
Ring embedding	[P]	Show ring arises from Y-junction topology
$T = \sigma/R$ identification	[Dc]	Explicit dimensional reduction with warp factor
Identical anchors $\lambda = \tau_n$	[Dc]	Prove from $\mathbb{Z}_N$ symmetry of brane
$W(u) \approx u$ linear form	[Dc]	Derive from GHY + Israel terms
Fourier restriction	[Der]	(Already derived)
$a_1 \propto 1/N$ from E-L	[Der]	(Already derived)
$k(N) = 1 + 1/N$	[Der]	(Already derived)

Summary: The toy functional form is *motivated* from 5D physics but not *derived* from first principles. The key [Dc] assumptions are:

1. Ring embedding exists and is $\mathbb{Z}_N$-symmetric
2. Brane deformation gives gradient energy $\propto (u')^2$
3. Localized defects at $\mathbb{Z}_N$ points give discrete anchors with identical λ

Once these assumptions are granted, the rest follows by standard variational mechanics [Der].

9 Conclusion

We have established the explicit mapping:

$$\underbrace{S_{5D}}_{\text{5D brane-world}} \xrightarrow{[Dc]} \underbrace{E[u] = \frac{T}{2}\int (u')^2 d\theta + \lambda \sum_n W(u_n)}_{\text{toy functional}} \xrightarrow{[Der]} \underbrace{a/c = 1/N}_{\text{normalization}} \tag{23}$$

What this upgrades:

- The toy functional is no longer an *ad hoc* construct
- Its form has a clear physical interpretation from 5D brane physics
- The "equal corner share" hypothesis ($a/c = 1/N$) has a derivation chain back to 5D

What remains open:

- Full derivation of ring embedding from Y-junction topology
- Explicit calculation of T and λ from 5D parameters
- BVP verification of $\mathbb{Z}_N$ mode structure